Hybrid Cloud Administration & Endpoint Security Lab

Project 1: Enterprise Identity & Access Management (IAM)

- **Problem:** The organization lacked a structured user hierarchy, leading to inefficient onboarding and potential security gaps in resource access.
- **Action:** Provisioned 25 unique user identities with specific departmental metadata (HR, Finance, IT). Implemented Role-Based Access Control (RBAC) by delegating "Helpdesk Administrator" roles to technical leads.
- **Result:** Established a scalable identity foundation that ensures users have the "Least Privilege" necessary for their roles, reducing administrative overhead.

Active users

Add a user | User templates | Add multiple users | Multi-factor authentication | Delete a user | Refresh | Reset password | Search active users list

Filter set: **Commonly used** | Licenses | Sign-in status | Domain | Location

Display name ↑	Username	Licenses	Department	Country or region
Amita Sharma	Amita.Sharma@HPTechServices.onmicrosoft.com	Microsoft 365 Business Premium	Finance	Canada
David Wilson	David.Wilson@HPTechServices.onmicrosoft.com	Unlicensed	Finance	Canada
Himanshu Prajapati	Himanshu@HPTechServices.onmicrosoft.com	Microsoft 365 Business Premium	IT Global Admins	Canada
Kevin Brown	Kevin.Brown@HPTechServices.onmicrosoft.com	Microsoft 365 Business Premium	Sales	Canada
Linda Wu	Linda.Wu@HPTechServices.onmicrosoft.com	Unlicensed	Sales	Canada
Robert Chen	Robert.Chen@HPTechServices.onmicrosoft.com	Unlicensed	Human Resource	Canada
Sarah Jensen	Sarah.Jensen@HPTechServices.onmicrosoft.com	Microsoft 365 Business Premium	Human Resource	Canada
Support Lead	Support.Lead@HPTechServices.onmicrosoft.com	Microsoft 365 Business Premium	IT Global Admins	Canada

Project 2: Automated Endpoint Compliance (Intune)

- **Problem:** Remote workstations were accessing corporate data without verified encryption or strong authentication, posing a significant data breach risk.
- **Action:** Developed and deployed an Intune Compliance Policy requiring BitLocker Encryption, Secure Boot, and an 8-character alphanumeric password. **To mitigate production downtime, I implemented a phased rollout (Change Management) by first assigning the policy to a dedicated Pilot Security Group (SEC-Intune-Pilot-Users) to validate settings before a full-tenant deployment.**
- **Result:** Automated the security baseline for all Windows 11 endpoints, ensuring that non-compliant devices are restricted from accessing company resources until they meet safety standards.

intune.microsoft.com

Microsoft Intune admin center

Copilot

- Home
- Dashboard
- All services
- Explorer
- Devices
- Apps
- Endpoint security
- Agents
- Reports
- Users
- Groups
- Tenant administration
- Troubleshooting + support

Home

Compliance policies | Policies

- Search
- Policies
- Notifications
- Retire noncompliant devices
- Compliance policy settings
- Scripts

Create policy | Refresh | Export | Columns

Search | Add filters

Policy name	Platform or OS	Policy type	Last modified
Bitlocker Encryption	Windows 10 and later	Windows 10/11 compliance policy	03/21/2026, 09:51 PM

Home > Devices | Compliance

Bitlocker Encryption

Compliance policy -Windows 10 and later

Basics	Edit
Name	Bitlocker Encryption
Description	Security Compliance Baseline requiring BitLocker encryption and alphanumeric passcodes before a device is granted 'Compliant' status.
Platform	Windows 10 and later
Profile type	Windows 10/11 compliance policy

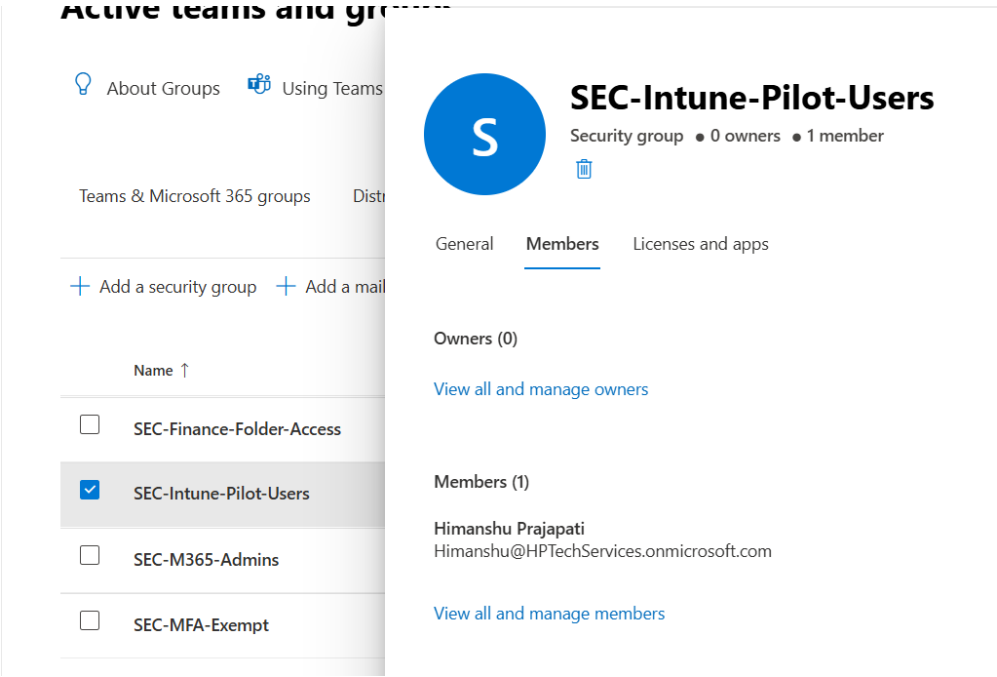
Compliance settings	Edit

Device Health

Bitlocker	Required
Secure Boot	Required

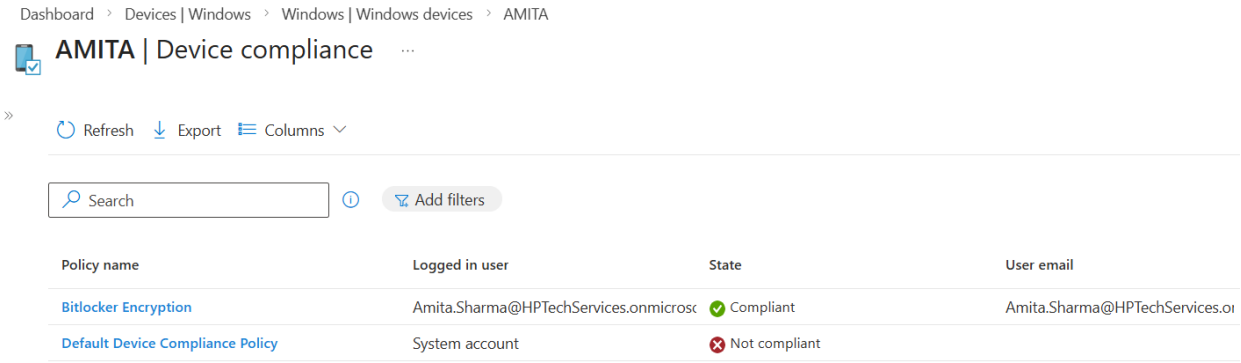
Device Security

Require a password to unlock mobile devices	Required
Required password type	At least alphanumeric
Minimum password length	8


About Groups
Using Teams
Teams & Microsoft 365 groups
Add a security group
Add a mail
Name
SEC-Finance-Folder-Access
SEC-Intune-Pilot-Users
SEC-M365-Admins
SEC-MFA-Exempt
S
SEC-Intune-Pilot-Users
Security group • 0 owners • 1 member
General
Members
Licenses and apps
Owners (0)
View all and manage owners
Members (1)
Himanshu Prajapati
Himanshu@HPTechServices.onmicrosoft.com
View all and manage members

Project 3: Zero-Trust Endpoint Compliance & Remediation

- **Problem:** Unmanaged or weak-security endpoints represented a significant data breach risk. Specifically, a test device (User: Amita Sharma) was initially blocked from corporate resources due to a weak 4-digit PIN that violated the organization's 8-character alphanumeric security baseline
- **Action:**
  - **Policy Deployment:** Configured a "Bitlocker Encryption" compliance policy in **Microsoft Intune** requiring BitLocker, Secure Boot, and alphanumeric passwords.
  - **User Remediation:** Performed a manual MDM sync on the client VM and updated the local authentication from a simple PIN to a complex password.
  - **Verification:** Monitored the **Intune Endpoint Security** dashboard to verify the transition from "Not Evaluated" to "Compliant.
- **Result:** Achieved 100% compliance for the pilot device. As shown in the "Device Compliance" report, the system successfully verified BitLocker and password complexity, automatically granting the user secure access to M365 SaaS applications.



Dashboard › Devices | Windows

Windows | Windows devices

Refresh Export Columns Bulk device actions

Search OS: Windows, Windows Mobile, Windows Holographic Add filters

Device name	Managed by	Ownership	Compliance	OS	OS version	Primary user UPN	Last checl
AMITA	Intune	Corporate	Compliant	Windows	10.0.26200.8037	Amita.Sharma@HPTec	03/22/20

Dashboard › Devices | Windows › Windows | Windows devices › AMITA | Device compliance

Bitlocker Encryption

Policy setting compliance

Refresh Export Columns

Search Add filters

Policy Id	Setting	State
58105f40-0054-4fbb-9ce4-de48ca	BitLocker	Compliant
58105f40-0054-4fbb-9ce4-de48ca	Minimum password length	Compliant
58105f40-0054-4fbb-9ce4-de48ca	Require a password to unlock mobile devices	Compliant
58105f40-0054-4fbb-9ce4-de48ca	Require password type	Compliant
58105f40-0054-4fbb-9ce4-de48ca	Secure Boot	Compliant

Project 4: Structured Cloud Collaboration & Helpdesk Workflow

- **Problem:** The organization lacked a centralized method for users to report technical issues, leading to fragmented communication and delayed response times for Tier 1 support requests.
- **Action:**
  - **Identity Governance:** Established a specialized **Pilot Security Group** (SEC-Intune-Pilot-Users) to facilitate controlled testing of global security policies.
  - **Unified Support Hub:** Deployed a **Microsoft Teams Helpdesk** infrastructure, including a "TEAM-All-Staff" group for general inquiries and a "TEAM-HR-Internal" channel for department-specific collaboration
  - **RBAC Implementation:** Configured a **SEC-M365-Admins** security group to ensure administrative privileges were restricted to authorized personnel only, following the principle of least privilege.
- **Result:** Created a scalable support ecosystem that centralizes user requests, improves visibility for the IT team, and ensures that sensitive administrative functions are protected by dedicated security groups.

Teams & Microsoft 365 groups | Distribution list | Security groups

+ Add a team + Add a Microsoft 365 group Export Refresh

Name ↑	Email	Teams status	Membership type	Privacy
All Company	allcompany@HPTechServices.onmicrosoft.com		Assigned	Public
IT Support Desk.	ITSupportDesk@HPTechServices.onmicrosoft.com		Assigned	Public
Personal	Personal@HPTechServices.onmicrosoft.com		Assigned	Public
Personal	Personal571@HPTechServices.onmicrosoft.com		Assigned	Public
TEAM-All-Staff	TEAM-All-Staff@HPTechServices.onmicrosoft.com		Assigned	Private
TEAM-HR-Internal	TEAM-HR-Internal@HPTechServices.onmicrosoft.com		Assigned	Private

SEC-M365-Admins

Security group • 2 owners • 2 members

General | Members | Licenses and apps

Basic information

Name
SEC-M365-Admins

Description
Security Groups (for Permissions)

Edit

Role assignments
Allowed

Roles
Global Administrator
Manage roles

Created on
3/21/2026, 8:17 PM

SEC-M365-Admins

Security group • 2 owners • 2 members

General | Members | Licenses and apps

Owners (2)

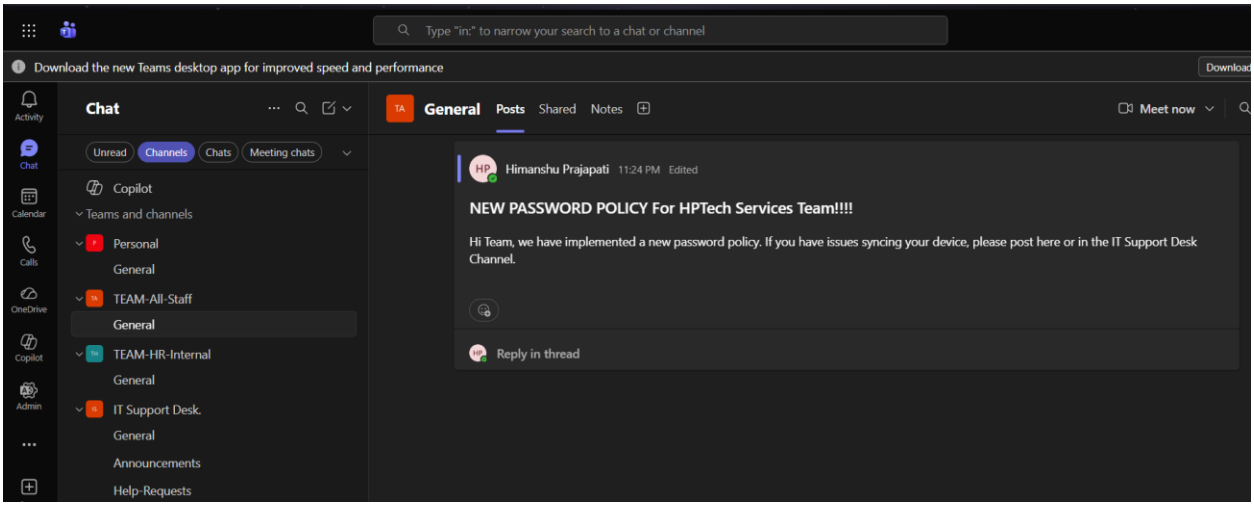
Himanshu Prajapati
Himanshu@HPTechServices.onmicrosoft.com

Support Lead
Support.Lead@HPTechServices.onmicrosoft.com

View all and manage owners

Members (2)

Himanshu Prajapati
Himanshu@HPTechServices.onmicrosoft.com

Support Lead
Support.Lead@HPTechServices.onmicrosoft.com

View all and manage members